

Quantifying relationships among populations

Simple tests

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Relationships among populations

Principal Component Analysis

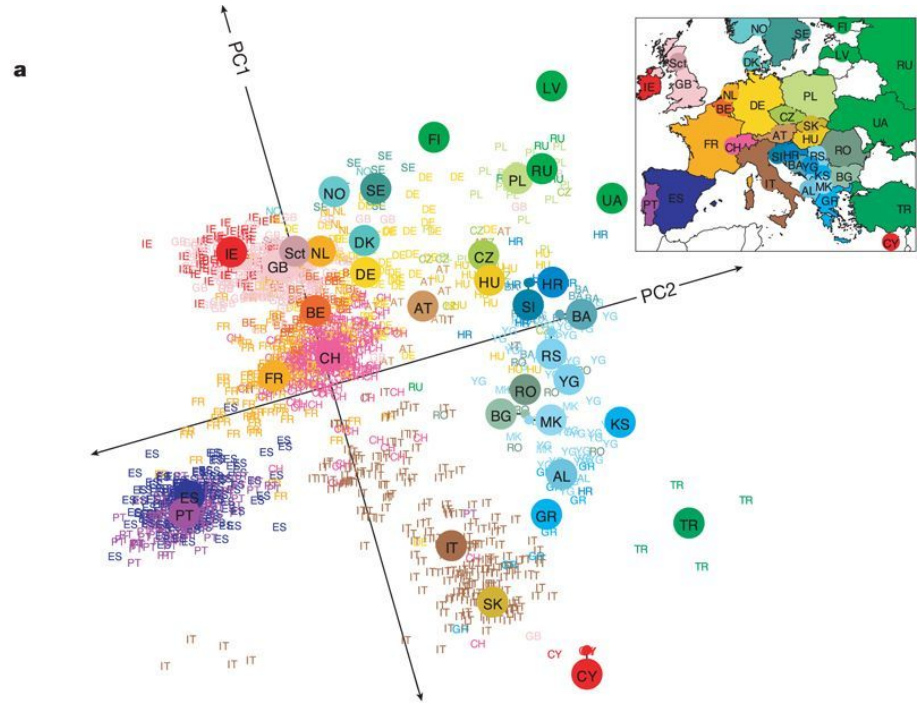
Genetic variation from multiple loci



Reduction of dimensions into
principal components (PCs)



Projection in space: proximity
indicates genetic similarity



Relationships among populations

ADMIXTURE Analysis

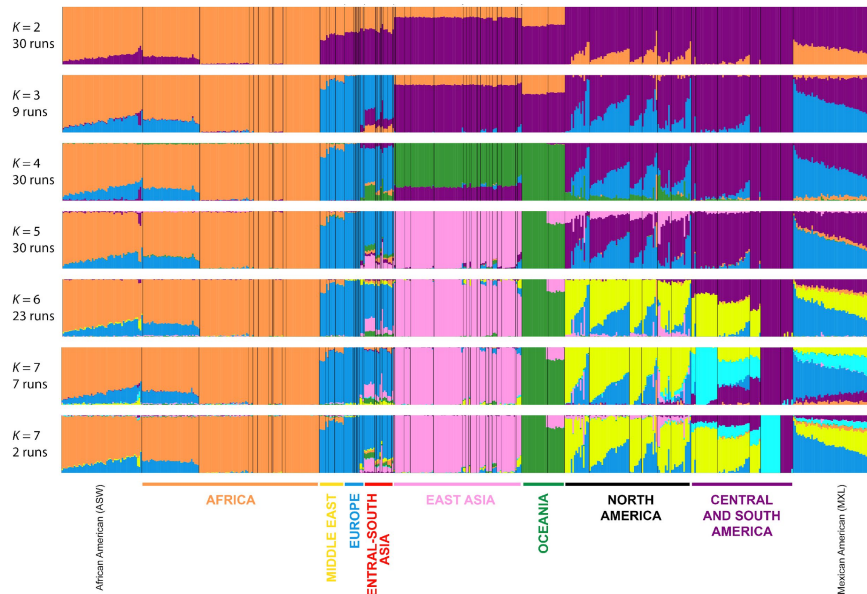
Genetic variation from multiple loci



Assumes there are k (2, 3, 4 ...) ancestry components



Describes each individual as a mixture of components



Admixture? Maybe not!

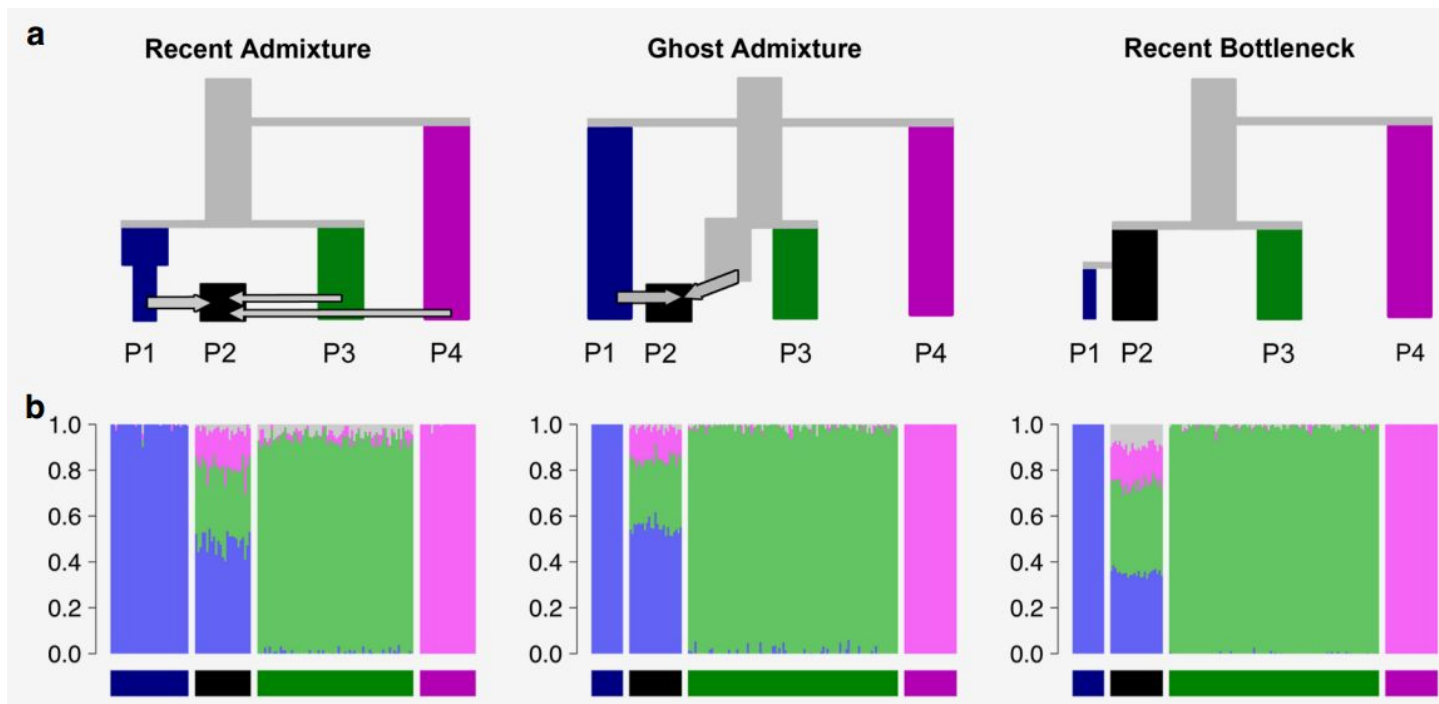
ARTICLE

DOI: 10.1038/s41467-018-05257-7

OPEN

A tutorial on how not to over-interpret
STRUCTURE and ADMIXTURE bar plots

Daniel J. Lawson¹, Lucy van Dorp^{2,3} & Daniel Falush⁴



F-statistics

Patterson et al., 2012 (*Genetics*)

Reich et al., 2009 (*Nature*)

Ancient Admixture in Human History

Nick Patterson,^{*,†} Priya Moorjani,[†] Yontao Luo,[‡] Swapan Mallick,[†] Nadin Rohland,[†] Yiping Zhan,[‡]

Teri Genschoreck,[‡] Teresa Webster,[‡] and David Reich^{*,†}

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ABSTRACT Population mixture is an important process in biology. We present a suite of methods for learning about population mixtures, implemented in a software package called ADMIXTOOLS, that support formal tests for whether mixture occurred and make it possible to infer proportions and dates of mixture. We also describe the development of a new single nucleotide polymorphism (SNP) array consisting of 629,433 sites with clearly documented ascertainment that was specifically designed for population genetic analyses and that we genotyped in 934 individuals from 53 diverse populations. To illustrate the methods, we give a number of examples that provide new insights about the history of human admixture. The most striking finding is a clear signal of admixture into northern Europe, with one ancestral population related to present-day Basques and Sardinians and the other related to present-day populations of northeast Asia and the Americas. This likely reflects a history of admixture between Neolithic migrants and the indigenous Mesolithic population of Europe, consistent with recent analyses of ancient bones from Sweden and the sequencing of the genome of the Tyrolean "Iceman."

F-statistics

Measure patterns of allele frequency correlations across populations

- Quantify shared drift (population similarity)
- Formal test of admixture
- Test models of population relationships

Advantages

- simple, flexible and fast compared to other popgen methods
- power to detect old admixture events
- “robust to all but the most extreme ascertainment processes”

F-statistics

F_2 statistics

Outgroup F_3

Admixture F_3 statistics

D statistics

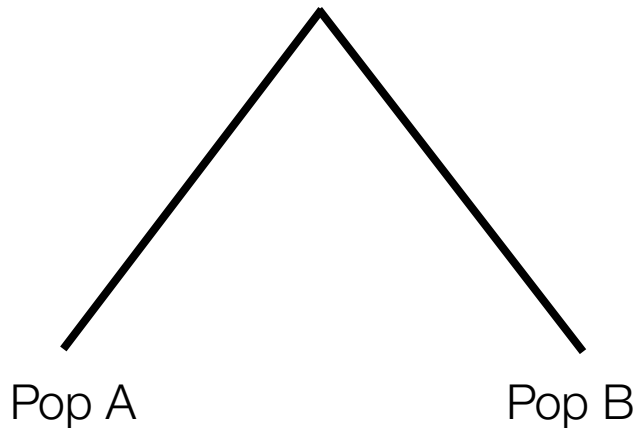
F_4 statistics

F_4 -ratio

How divergent are two populations?

$$F_2(A, B) = \mathbb{E}[(p_A - p_B)^2] = \mathbb{E}[(p_A - p_B)(p_A - p_B)]$$

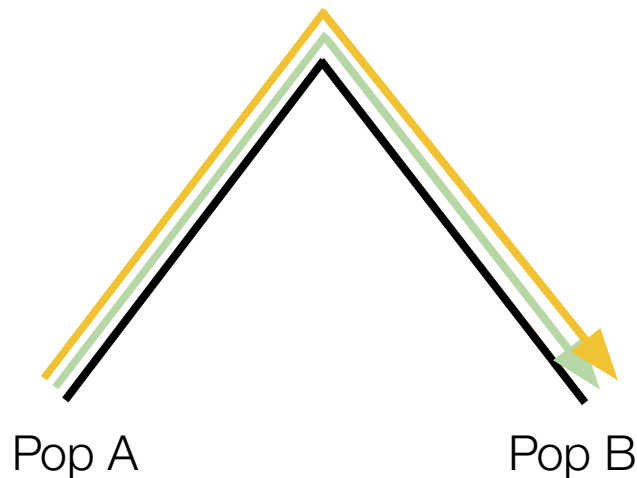
Average squared difference of allele frequencies



How divergent are two populations?

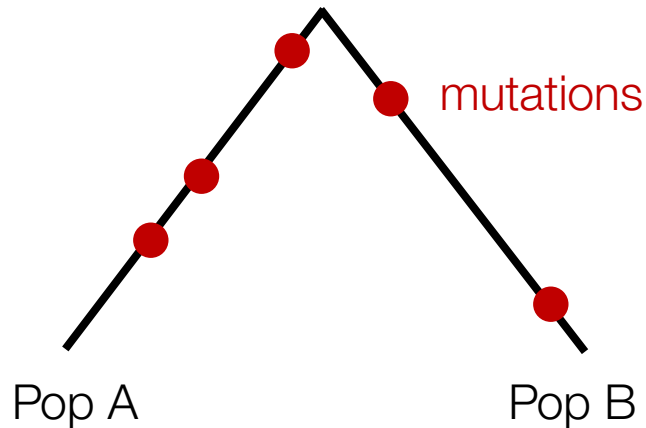
$$F_2(A, B) = \mathbb{E}[(p_A - p_B)^2] = \mathbb{E}[\underbrace{(p_A - p_B)}_{\text{yellow arrow}} \underbrace{(p_A - p_B)}_{\text{green arrow}}]$$

Average squared difference of allele frequencies



How divergent are two populations?

$$F_2(A, B) = \mathbb{E}[(p_A - p_B)^2] = \frac{1}{L} \sum_{l=1}^L (p_A(l) - p_B(l))^2$$

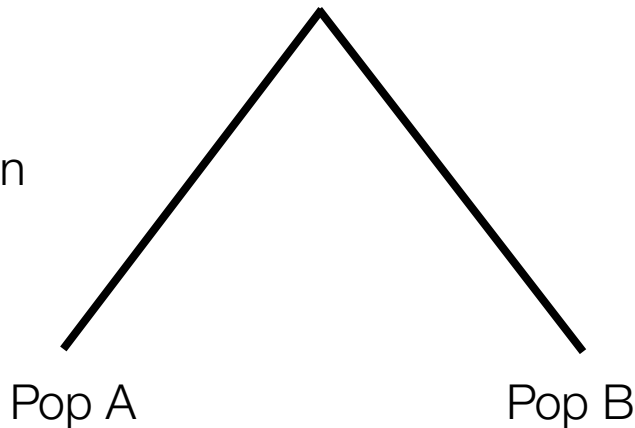


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Assume we sample 1 chromosome per population
(p_A and p_B only take values 0 or 1) and $A = B$?

$$F_2(A, B) = 0$$

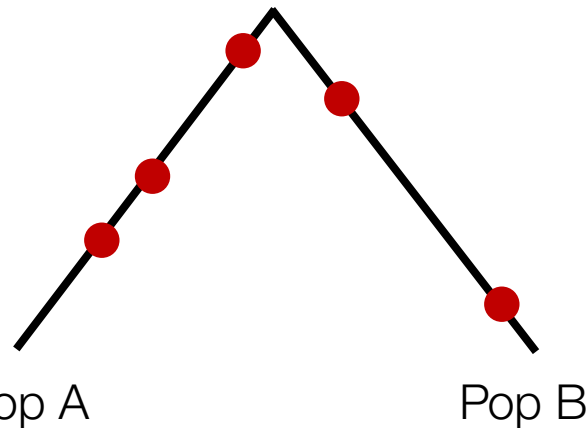


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Assume we sample 1 chromosome per population (p_A and p_B only take values 0 or 1) and $A \neq B$?

$F_2(A, B)$ = mean number of pairwise differences

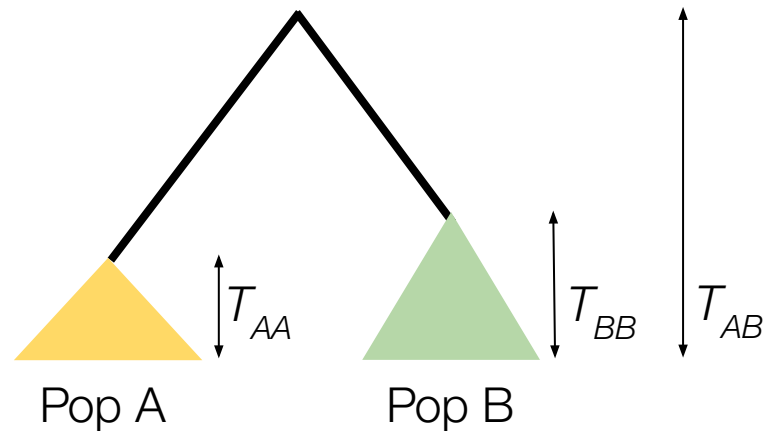


How divergent are two populations?

$$F_2(A, B) = 2\mu \left(\mathbb{E}[T_{AB}] - \frac{\mathbb{E}[T_{AA}] + \mathbb{E}[T_{BB}]}{2} \right)$$

Alternative formulation, where T_{AB} is the time to the most recent common ancestor (TMRCA) of A and B.

If A and B have low within-population diversity, T_{AA} and T_{BB} are very small and $F_2(A, B)$ is larger.

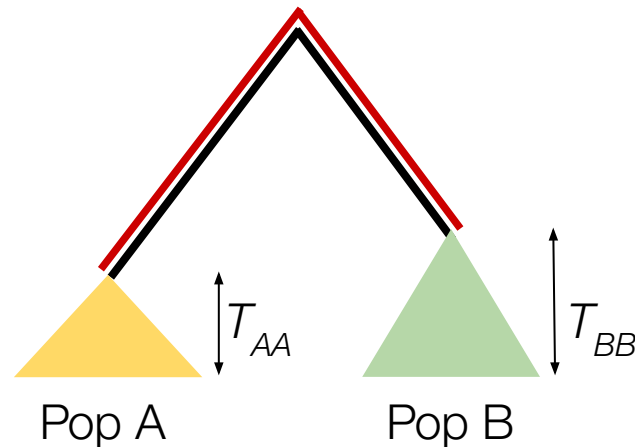


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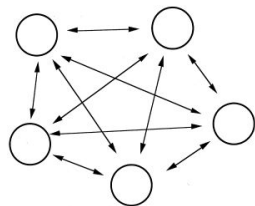
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If A and B have low within-population diversity, T_{AA} and T_{BB} are very small and $F_2(A, B)$ is larger.



How divergent are two (or more) populations?

F_{ST} is similar to F_2 but has a normalization that constrains it between 0 and 1



H_T = total heterozygosity (assuming all sub-populations come from one group)

H_S = average heterozygosity within sub-populations (average of individual H values)

$$F_{ST} = (H_T - H_S) / H_T$$

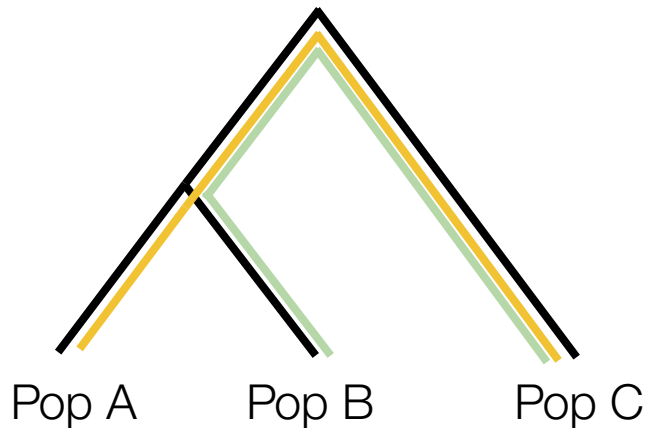
(the proportion of the total variance due to between-population differences)

Shared drift between A and B (outgroup F₃)

$$F_3(C; A, B) = \mathbb{E}[(\underline{p_C - p_A})(\underline{p_C - p_B})] = \frac{1}{L} \sum_{l=1}^L (p_C(l) - p_A(l))(p_C(l) - p_B(l))$$

Shared path from pop C to pop A and pop C to pop B

No dependency on T_{AA} and T_{BB}

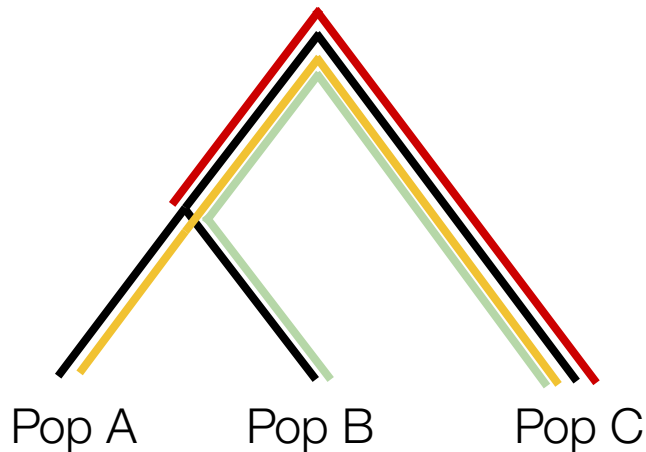


Shared drift between A and B (outgroup F_3)

$$\underline{F_3(C; A, B)} = \mathbb{E}[(\underline{p_C - p_A})(\underline{p_C - p_B})] = \frac{1}{L} \sum_{l=1}^L (p_C(l) - p_A(l))(p_C(l) - p_B(l))$$

Shared path from pop C to pop A and pop C to pop B

No dependency on T_{AA} and T_{BB}



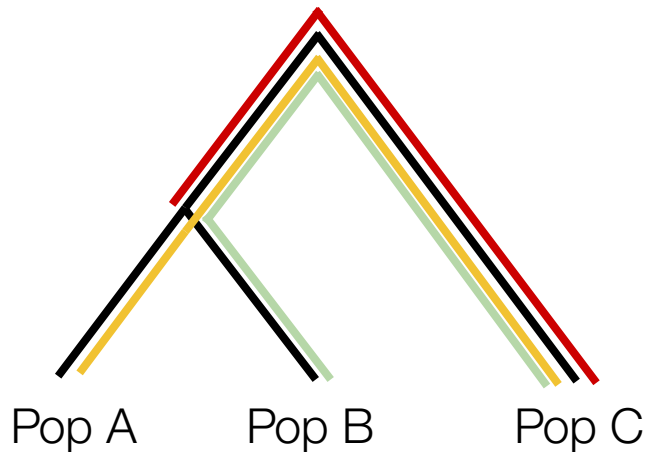
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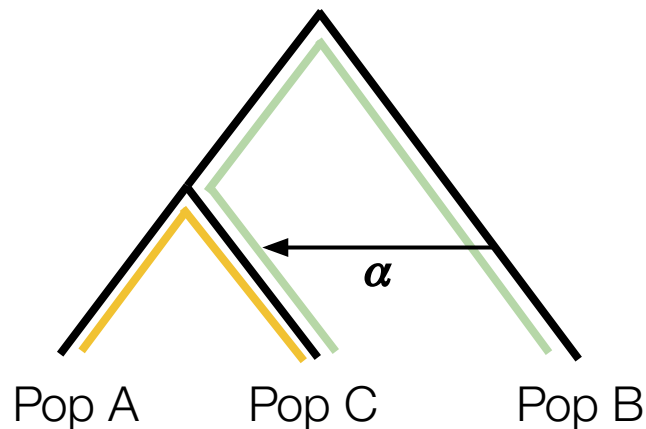
Measures similarity: the highest the F_3 value, the closest pop A and pop B!



Admixture F_3 (C as admixture target)

$$F_3(C; A, B) = \mathbb{E}[(\underline{p_C - p_A})(\underline{p_C - p_B})] = \frac{1}{L} \sum_{l=1}^L (p_C(l) - p_A(l))(p_C(l) - p_B(l))$$

Shared path from pop C to pop A and pop C to pop B

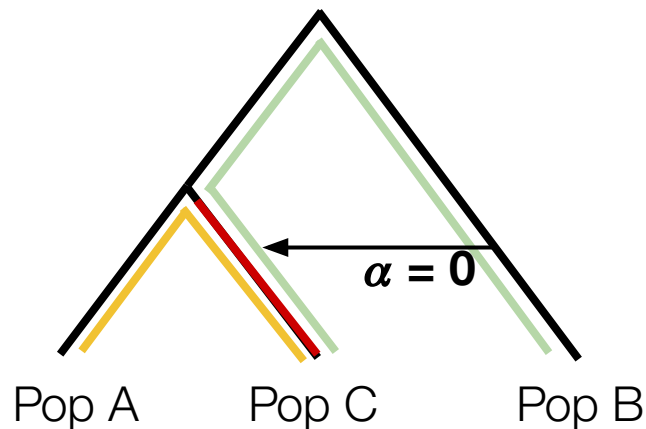


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Shared path from pop C to pop A and pop C to pop B

$F_3 > 0$ if $\alpha = 0$

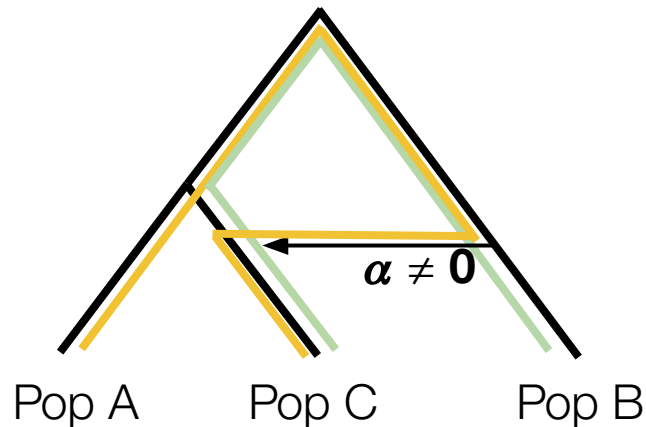


Admixture F_3 (C as admixture target)

$$F_3(C; A, B) = \mathbb{E}[(\underline{p_C - p_A})(\underline{p_C - p_B})] = \frac{1}{L} \sum_{l=1}^L (p_C(l) - p_A(l))(p_C(l) - p_B(l))$$

Shared path from pop C to pop A and pop C to pop B

$F_3 < 0$ (negative) only if $0 < \alpha < 1$



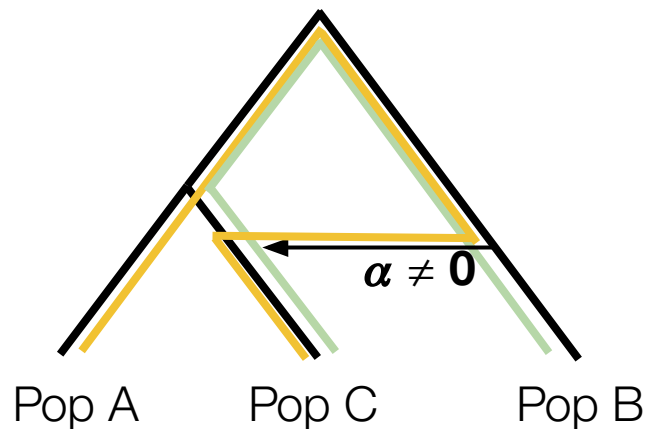
Admixture F_3 (C as admixture target)

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Shared path from pop C to pop A and pop C to pop B

But a positive F_3 does not guarantee that there is no admixture (test has low power if there is high drift)

Most useful for recent admixture at approximately equal proportions

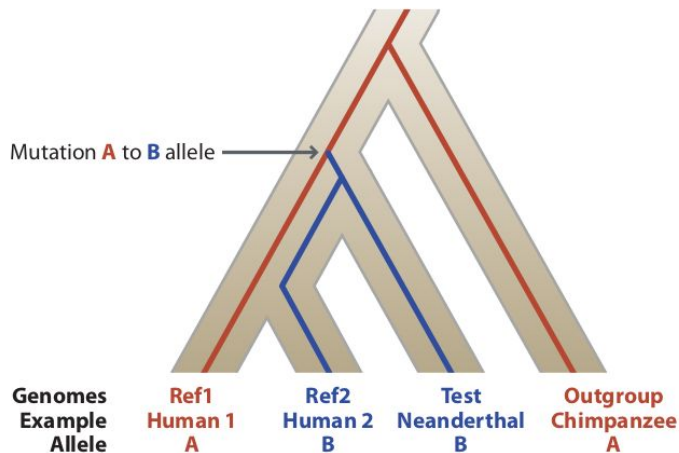


D statistic or ABBA BABA test

$$D = \frac{ABBA - BABA}{ABBA + BABA}$$

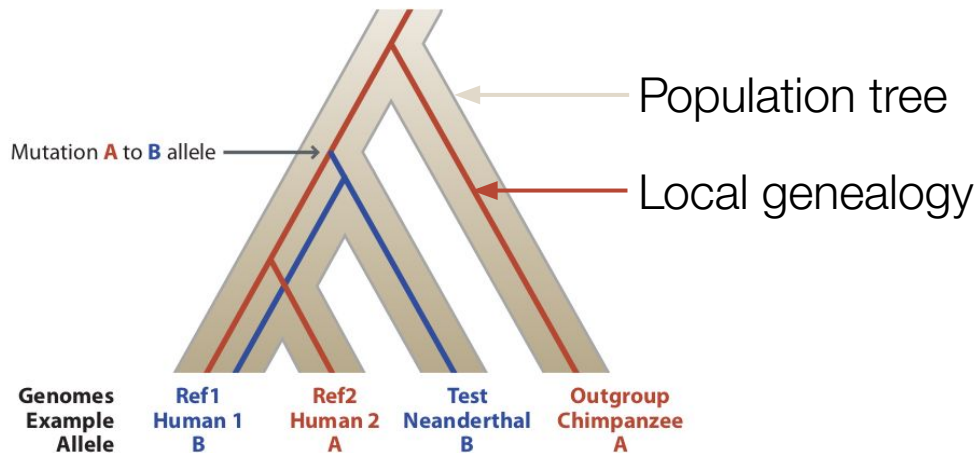
ABBA

Human 2 shares allele with Neanderthal
Increases genomewide value of Patterson's D



BABA

Human 1 shares allele with Neanderthal
Decreases genomewide value of Patterson's D



D statistic or ABBA BABA test

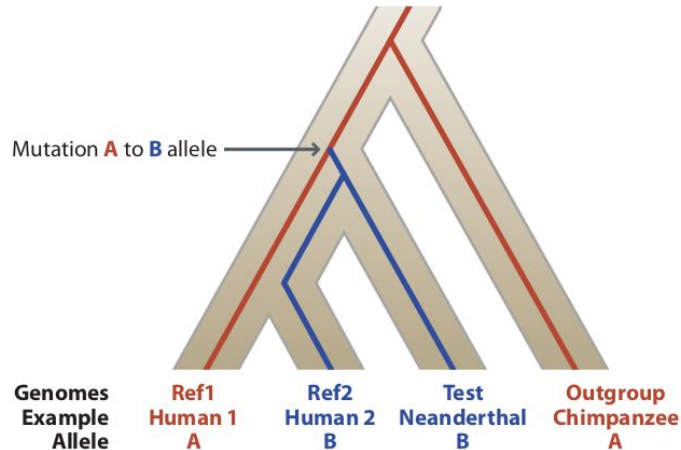
$$D = \frac{ABBA - BABA}{ABBA + BABA}$$

$D = 0$, if $ABBA = BABA$

$D \neq 0$, if $ABBA \neq BABA$

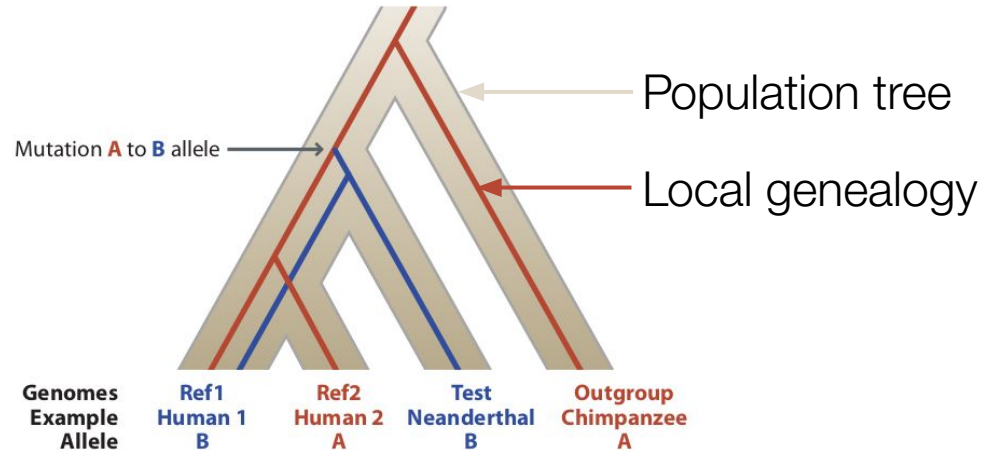
ABBA

Human 2 shares allele with Neanderthal
Increases genomewide value of Patterson's D



BABA

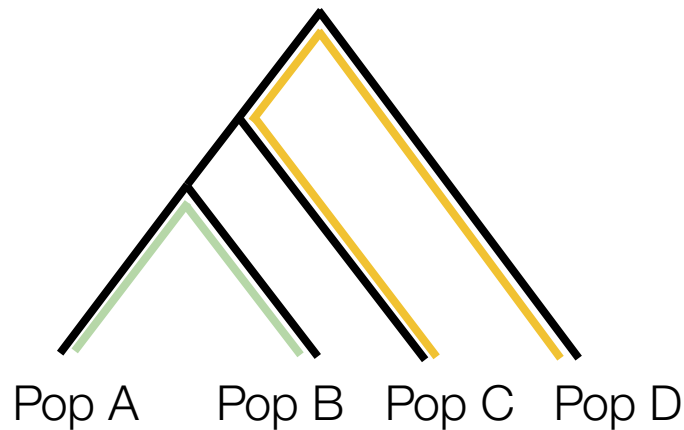
Human 1 shares allele with Neanderthal
Decreases genomewide value of Patterson's D



F_4 statistic

$$F_4(A, B; C, D) = \mathbb{E}[\underbrace{(p_A - p_B)}_{\text{green}} \underbrace{(p_C - p_D)}_{\text{yellow}}]$$

Covariance of allele frequencies between two pairs of populations (same as D-statistics, but using units of drift)

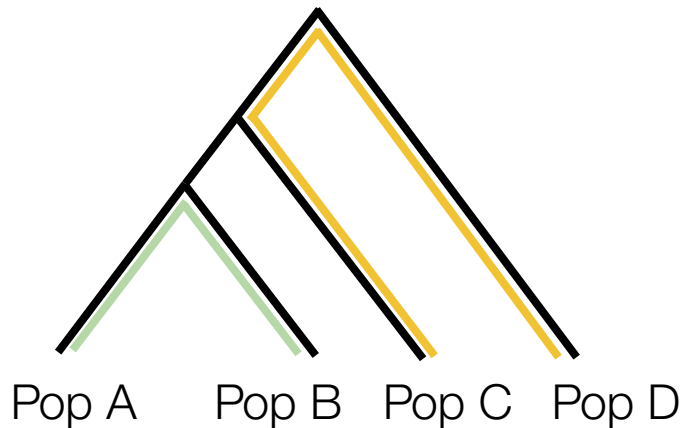


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If there is **no admixture**, there is no covariance between pairs and no overlap between the compared paths, therefore $F_4 = 0$

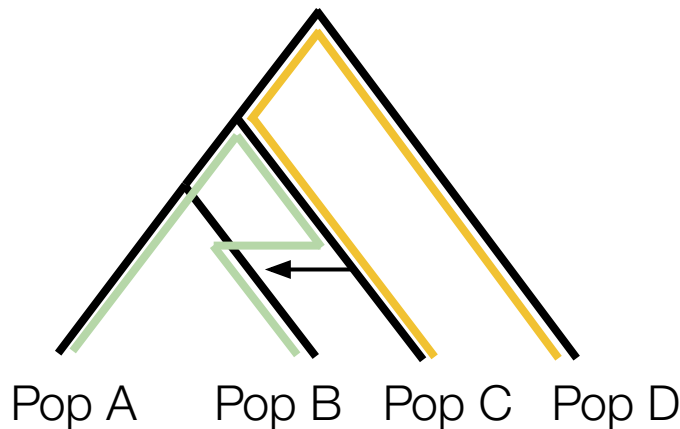


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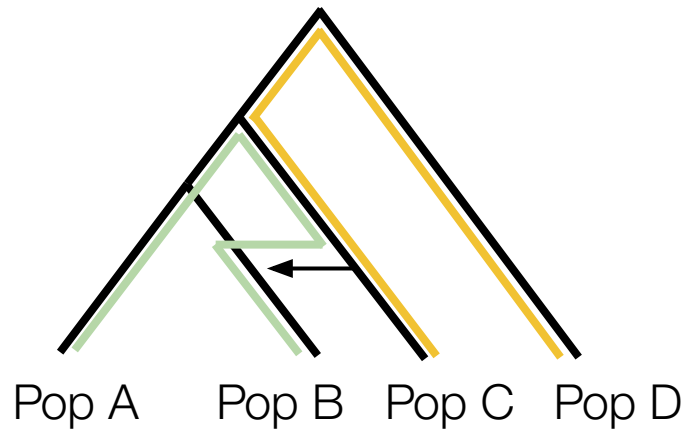
If there is **admixture**, there is some covariance between pairs and some overlap between the compared paths, therefore $F_4 \neq 0$



F_4 statistic

$$F_4(A, B; C, D) = \mathbb{E}[\underbrace{(p_A - p_B)}_{\text{green}} \underbrace{(p_C - p_D)}_{\text{yellow}}]$$

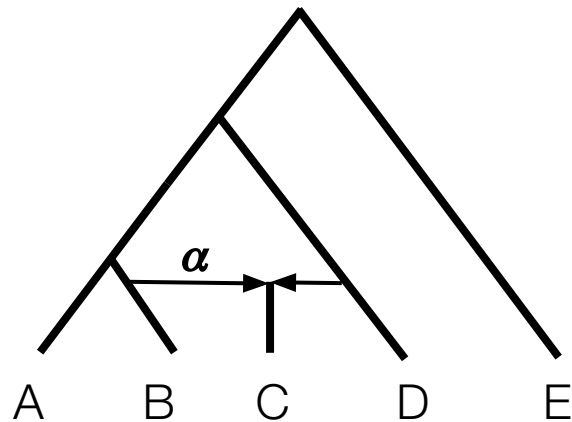
F_4 is more sensitive to admixture (including old admixture) than Admixture F_3



F_4 ratio

$$F_4 \text{ ratio} = \frac{F_4(A, E; D, C)}{F_4(A, E; D, B)} = \alpha$$

F_4 ratio allows estimating ancestry proportions in an admixed population, under the assumption that we have a correct historical model



Evaluating significance in F-statistics

Block jackknife

- the genome is divided into contiguous blocks of ~5 cM
- the statistic is recomputed leaving out each block in turns
- the variance across those estimates is converted to a standard error (SE)
- A z-score (estimate / SE) is reported; $|z| > 3$ is a commonly used significance threshold

Ascertainment bias in F-statistics

- in theory, SNPs should be ascertained in an outgroup to all tested populations
- common SNPs are older; f -statistics mostly reflect deep history
- the sign of the f_4 can flip under different ascertainment/SNP sets
- missing data compounds the problem
- f_4 more robust than f_2 , but not immune
- repeat analyses under different ascertainment schemes

F-statistics recap

Measure patterns of allele frequency correlations across populations

- Quantify shared drift (population similarity)
- Formal test of admixture
- Test models of population relationships

Advantages

- simple, flexible and fast compared to other popgen methods
- power to detect old admixture events
- “robust to all but the most extreme ascertainment processes”

Population genomics:
background and tools

15 – 22 June 2026 | Campania region, Italy

Next:
Practical using
admixtools v2